Pimpri Chinchwad Education Trust's

Pimpri Chinchwad College of Engineering

Department of Computer Engineering(Regional Language)

B.TECH MAJOR PROJECT PRESENTATION

INFRASYNC: INTELLIGENT & TRANSPARENT ROAD CONSTRUCTION

An AI and Blockchain-based framework for efficient and transparent road infrastructure management

Domain: Smart City | E-governance

Guide:

Mrs. Rucha Shinde

GA9:

Team Pixel Pioneers

Parag Pawar	**122B1D051**
Om Sagar	**122B1D055**
Harshal Suroshe	**122B1D061**
Vinayak Suryawanshi	**122B1D062**

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PROBLEM STATEMENT

- Road construction projects frequently face delays, cost overruns, and poor quality due to manual planning, inefficient monitoring, and lack of accurate cost and time prediction.

- Fund allocation and usage lack transparency and accountability, making it difficult for authorities to audit transactions and prevent misuse of public money.

- Citizens have limited visibility into project progress and no effective system to provide feedback, leading to reduced trust and unresolved infrastructure issues.

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40% of Indian road projects hit cost overrun bump: WB

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OBJECTIVES

- **Predict road construction cost and duration using Machine Learning**
- **Ensure transparent and tamper-proof fund tracking using Blockchain**
- **Integrate citizen feedback using NLP-based sentiment analysis**
- **Provide a single digital platform for government, contractors, and citizens**

SDG GOALS:

9. Industry, Innovvation and Infrastructure
11. Sustainable cities and communities

GAP IDENTIFICATION

What exists today ?

- E-procurement systems handle tenders but not execution transparency
- Project management tools track timelines but:
  Do not predict delays ; Do not track funds immutably
- Citizen complaint systems exist but are isolated and underutilized

Identified Gaps

- No predictive planning before project execution
- No immutable audit trail for fund allocation
- No integration of citizen sentiment with project performance
- Systems work separately, not as a unified platform

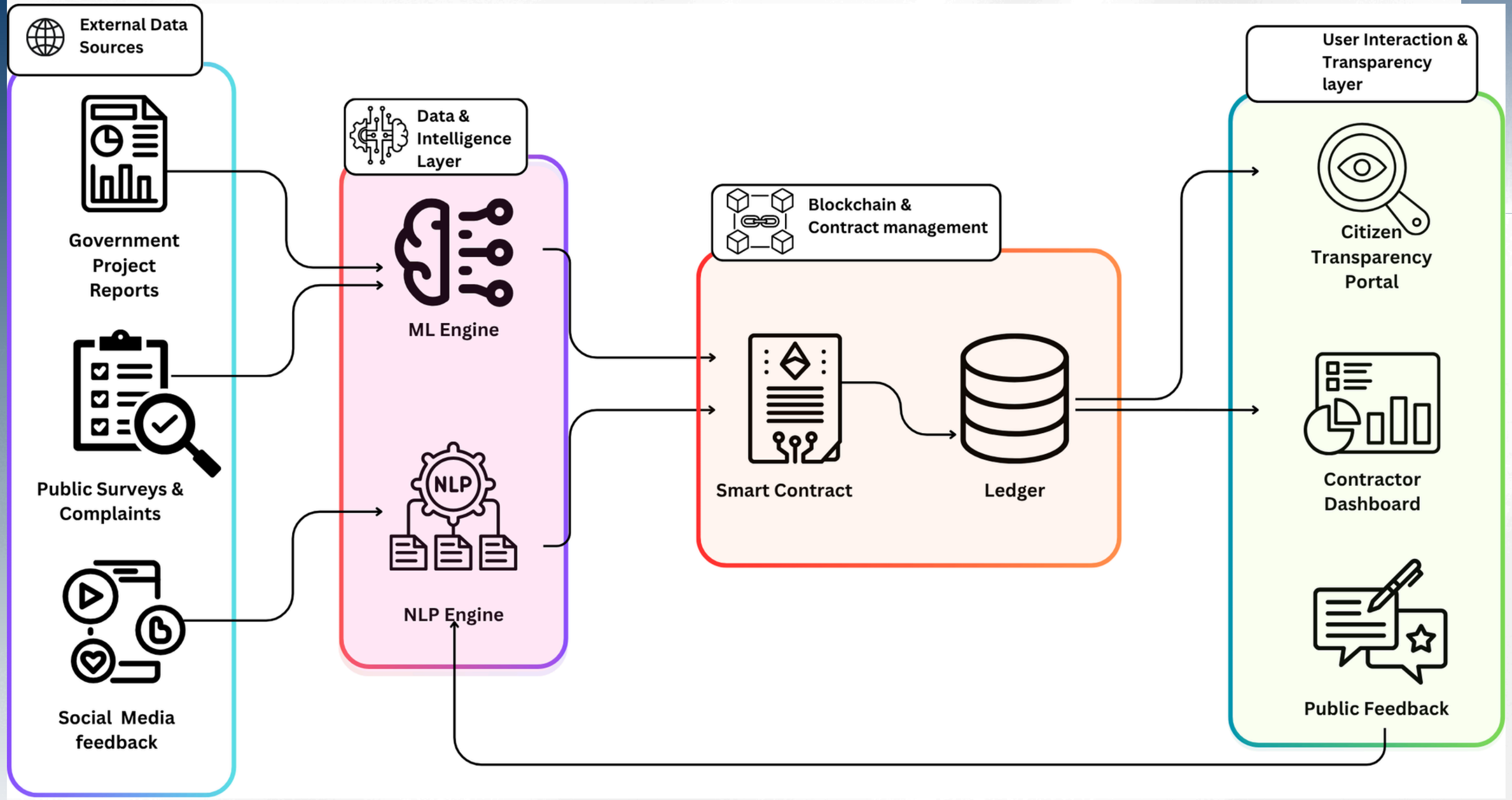
PROPOSED SOLUTION: ***INFRASYNC***

Our Approach

InfraSync is an integrated platform that combines:

- *Machine Learning* → Predict cost and time before execution
- *Blockchain* → Record fund transactions and approvals immutably
- *NLP (Sentiment Analysis)* → Analyze citizen feedback
- *Web Platform* → Role-based dashboards for all stakeholders

WORKFLOW


External Data Sources
Government Project Reports
Public Surveys & Complaints
Social Media feedback
Data & Intelligence Layer
ML Engine
NLP
NLP Engine
Blockchain & Contract management
Smart Contract
Ledger
User Interaction & Transparency layer
Citizen Transparency Portal
Contractor Dashboard
Public Feedback

WORKFLOW

Step-by-step flow:

1. Government enters project details

2. ML model predicts estimated cost and completion time

3. Project data and fund allocation are recorded on blockchain

4. Funds are released milestone-wise

5. Contractors update project progress

6. Citizens submit feedback

7. NLP analyzes sentiment (positive / negative / neutral)

8. Authorities get real-time insights and alerts

USECASES AND IMPACT

Who benefits?

Government:

Better budgeting, monitoring, and transparency

Contractors:

Clear fund release mechanism and planning support

Citizens:

Visibility into projects and ability to voice concerns

Impact:

- Better quality roads
- Efficient use of public money
- Increased public trust in government

CONCLUSION AND FUTURE SCOPE

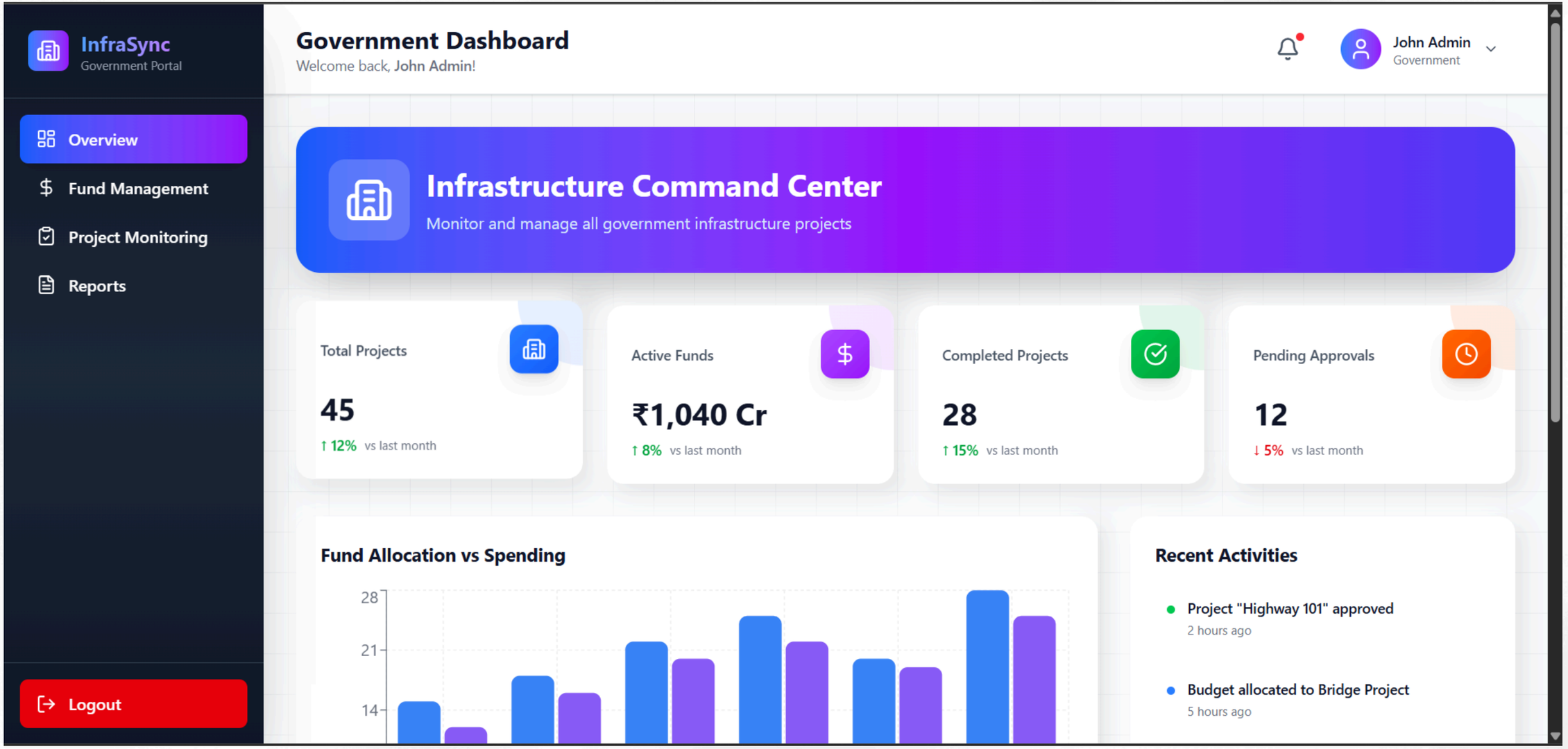
Conclusion:

InfraSync provides a practical and scalable solution to long-standing issues in road construction by combining AI-driven prediction, blockchain-backed transparency, and citizen feedback analysis.

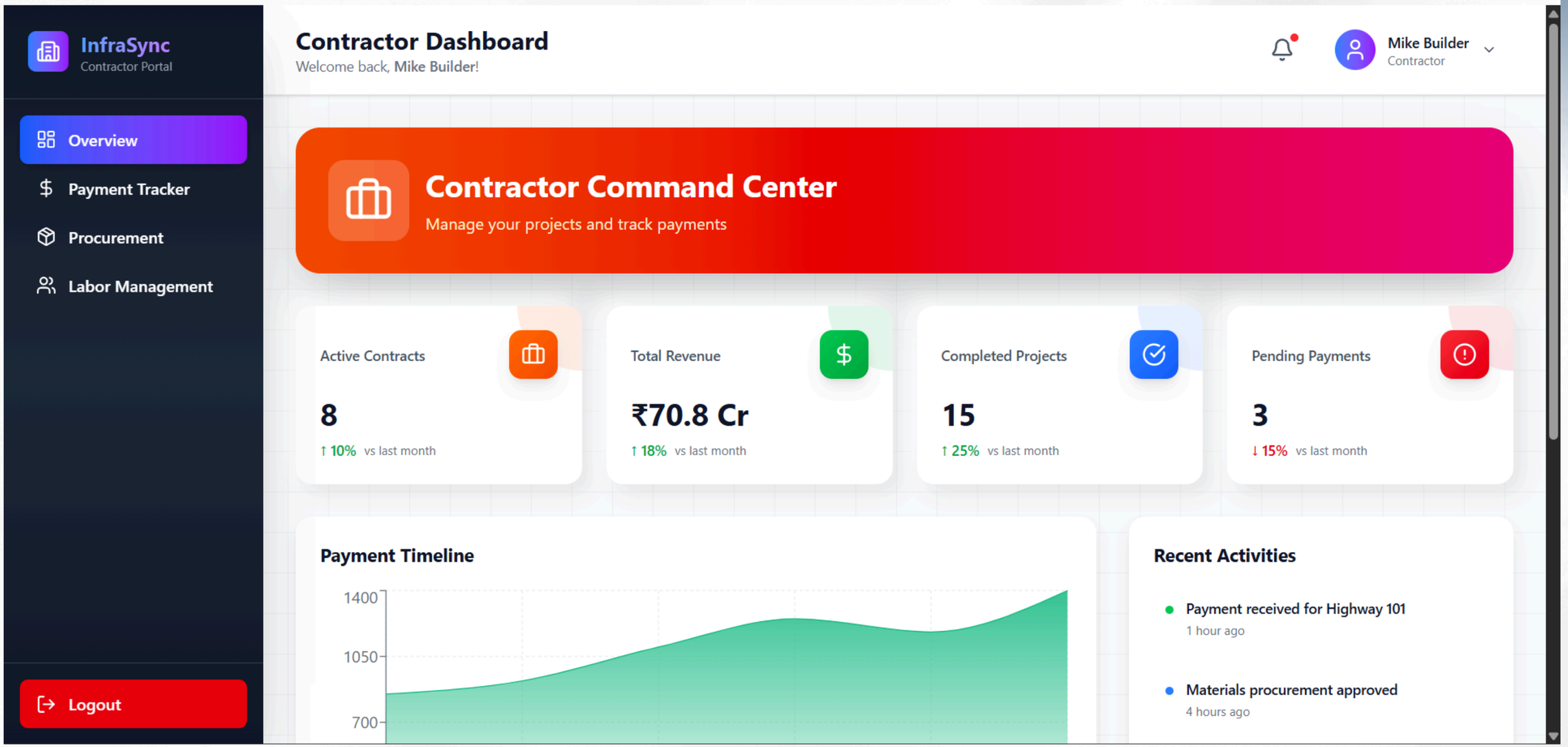
Future Scope:

- Expansion to other infrastructure projects (bridges, railways, housing)
- Integration with government finance systems
- Advanced fraud detection using AI
- Nationwide smart infrastructure monitoring
- Integration of sensors to determine material quantity

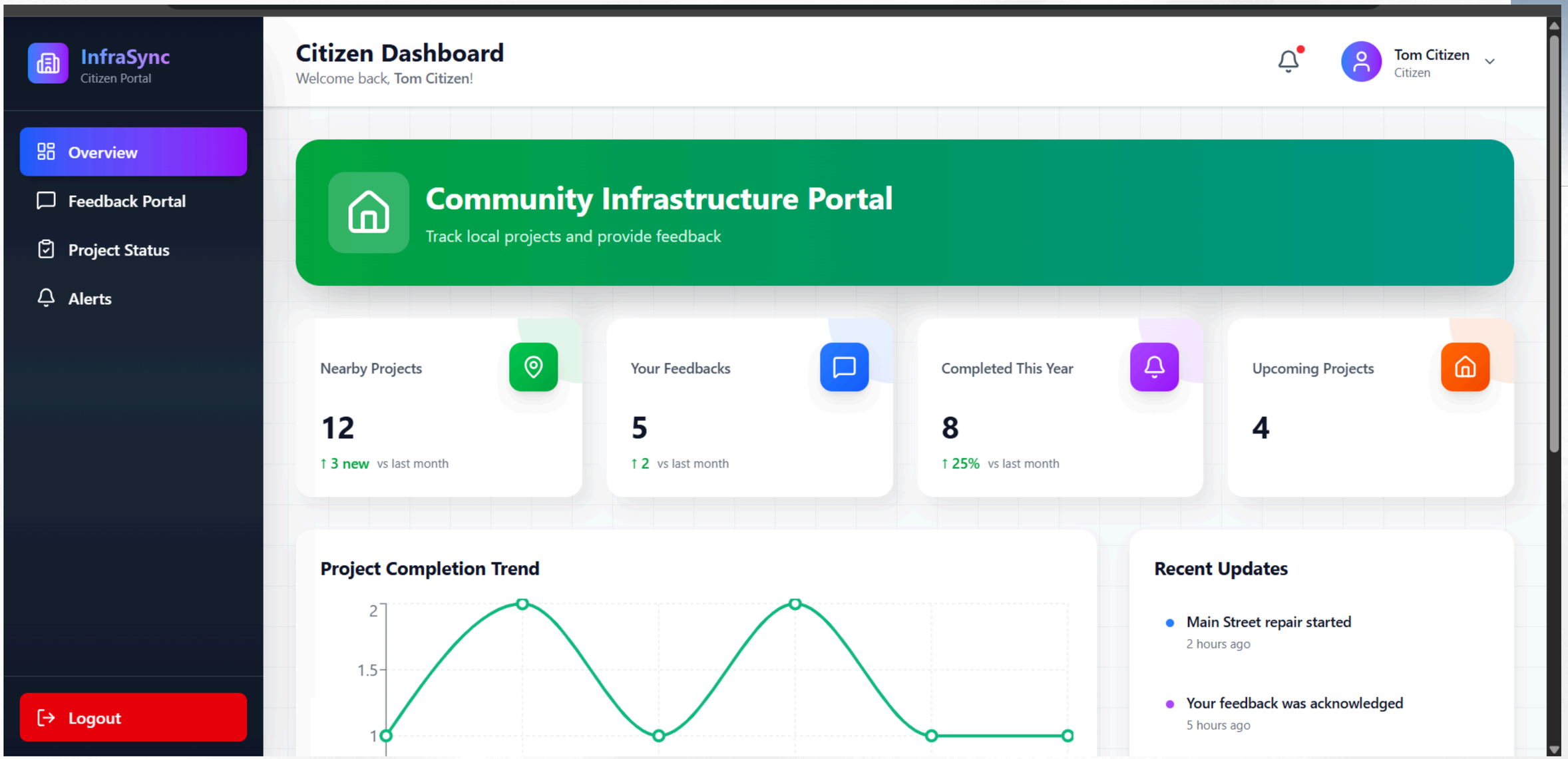
Government Interface/Dashboard



Contractor Interface/Dashboard



Citizen Interface/Dashboard



THANK YOU



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Project Title:

Infrasync: Intelligent and Transparent Road Construction

Guided By:
Mrs. Rucha Shinde

Group Members:

Parag Pawar	**122B1D051**
Om Sagar	**122B1D055**
Harshal Suroshe	**122B1D061**
Vinayak Suryawanshi	**122B1D062**

PROBLEM STATEMENT

Road construction projects frequently face delays, cost overruns, and poor quality due to manual planning, inefficient monitoring, and lack of accurate cost and time prediction

Fund allocation and usage lack transparency and accountability, making it difficult for authorities to audit transactions and prevent misuse of public money

Citizens have limited visibility into project progress and no effective system to provide feedback, leading to reduced trust and unresolved infrastructure issues.

OBJECTIVE

Predict & Prevent:
To predict the exact cost and timeline of road projects before they start using historical data.

100% Financial Transparency:
To make fund transfers immutable (unchangeable) so every rupee is tracked and accounted for.

Unified Governance:
To build a single, transparent portal where the government plans, contractors execute, and citizens monitor.

LITERATURE SURVEY

Title	Methodology	Algorithms Used	Research Gap	Future Scope
Low-Carbon Concrete with Blockchain & ML	Uses Blockchain for data security and ML for mixture optimization.	Random Forest, Gradient Boosting, Blockchain	Lacks large-scale real-world validation; complex integration.	Real-world application, improved scalability, deep learning integration.
Blockchain Smart Contracts in Construction	Analyzes adoption barriers and proposes an implementation roadmap.	Smart Contracts, Analytical Frameworks	Limited real-world use and poor integration with existing systems.	Pilot projects, IoT/AI integration, standardized frameworks.
Highway Cost Prediction using ML	Analyzes historical data with ML for accurate cost prediction.	Linear Regression, Random Forest, SVM	Accuracy relies on data quality; ignores economic fluctuations.	Larger datasets, deep learning, real-time data integration.
AI and Blockchain in Construction Mgmt.	Reviews AI & Blockchain integration for efficiency and transparency.	AI, ML, Blockchain	Limited practical use; no standardized integration framework.	Real-time integrated platforms, large-scale application.
Road Cost Estimation with Explainable ML	Predicts costs using Explainable ML for decision transparency.	XAI, Random Forest, Gradient Boosting, SHAP/LIME	Limited dataset diversity; poor geographic generalization.	Larger datasets, integrating economic and environmental factors.

PROPOSED SOLUTION

How our solution stands out:

We are moving away from standard Excel sheets and basic websites.



ML-Powered Estimation:

When the government enters project details, our system instantly predicts the required budget and timeline.



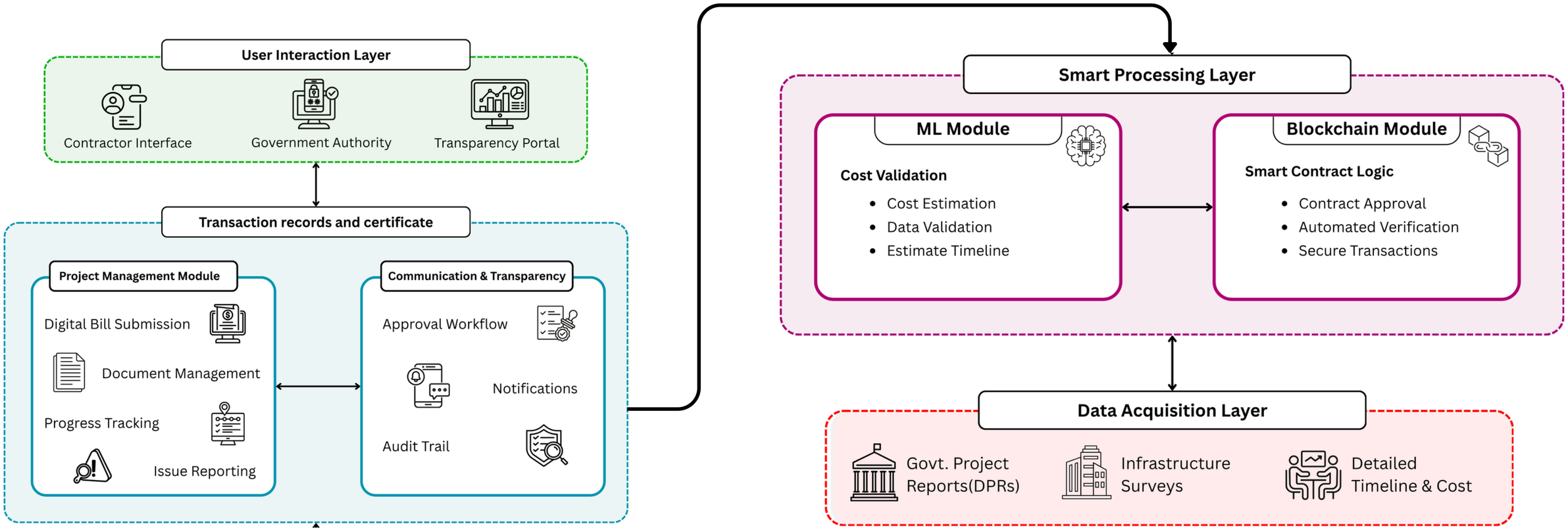
Milestone-Based Smart Payments:

Funds are not just transferred blindly. They are locked in a digital contract and released automatically only when contractors complete specific milestones.

The Result:

Faster project completion, zero financial tampering, and high public trust.

Architecture diagram



COMPLEXITY OF PROJECT

The project involves multiple layers and technologies, making it moderately complex.



It includes integration of:

- Frontend (Next-based UI)
- Backend (Fast APIs)
- Database
- Machine Learning models for prediction
- Blockchain-based smart contract simulation

Handling real-time updates, role-based access control, and secure transactions adds to the system complexity.
Additionally, preparing and calibrating datasets from government sources increases the technical depth of the project.

PROJECT PLAN

The project is executed in structured phases to ensure systematic development.

- Requirement Analysis: Understanding problem statement and collecting data
- System Design: Designing architecture, database schema, and UI
- Development: Implementing frontend and backend modules
- Blockchain Integration: Developing smart contract-based fund tracking
- Testing: Verifying functionality and fixing errors
- Documentation: Preparing reports and final presentation

The development follows a modular and incremental approach, ensuring smooth progress and easy debugging.

SOFTWARE TESTING

Testing plays a crucial role in ensuring system reliability.

Unit Testing

Individual modules like authentication, project creation, and milestone updates were tested separately

Ensured each component works independently without errors.

Integration Testing

Verified interaction between:

- Frontend ⇔ Backend APIs
- Backend ⇔ Database
- Backend ⇔ Smart Contract module

Ensured smooth data flow across the system.

API Testing

Tested all REST APIs such as:

- Login/Register
- Project creation
- Fund allocation & release
- Feedback submission

System Testing

Tested complete workflow:
Project creation → milestone update → fund release.

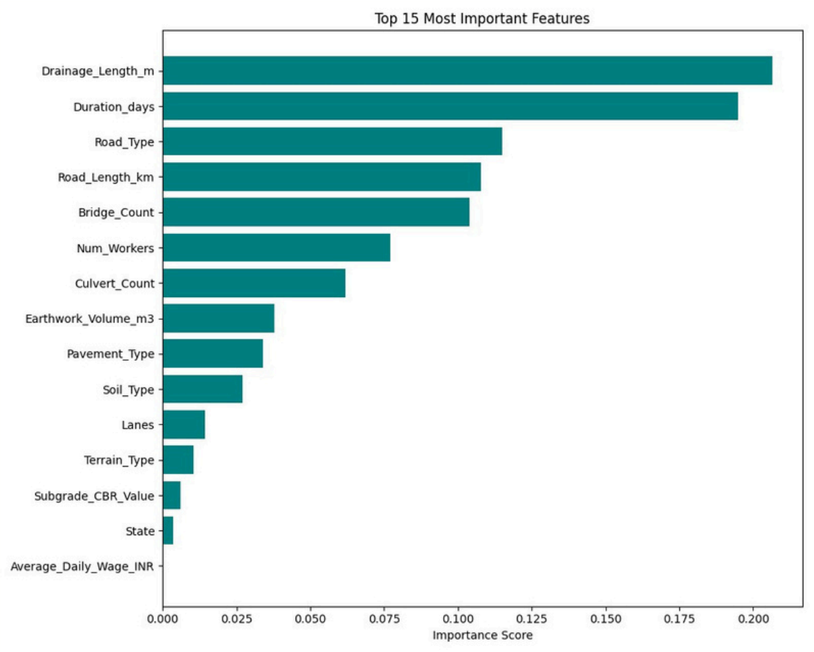
Verified role-based access for Government, Contractor, and Citizen

ALGORITHMS USED

Machine Learning (XGBoost)

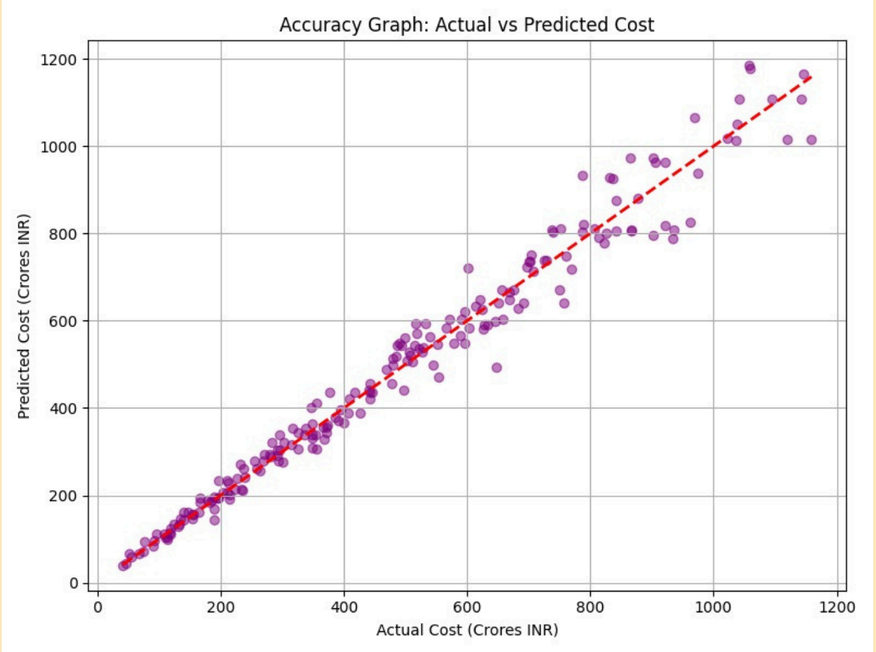
Machine learning algorithm used for predicting road construction cost.

- Data Sourcing: Extracting key features (road length, type, duration) from official Detailed Project Reports (DPRs).
- Preprocessing: Cleaning, encoding, and normalizing the raw data using Pandas.
- XGBoost Training: Building sequential decision trees that learn from past errors to maximize prediction accuracy.
- Live Prediction: Entering new project parameters into InfraSync to instantly generate data-driven cost estimates.



Feature Importance Graph:

- The graph indicates that Drainage Length, Project Duration, Road Type, and Road Length are major contributors to project cost estimation.



Actual vs Predicted Cost Graph:

- The scatter plot maps real project costs against our XGBoost predictions.
- The data points cluster tightly along this line, proving the model is highly reliable and accurate.(85%)

BLOCKCHAIN

Why Blockchain?

A standard database (like SQL or MongoDB) can be secretly edited by an administrator. This leads to corruption.

- We use Blockchain because it is immutable. Once a fund release is recorded, it is permanently locked. It can never be deleted, altered, or hidden, guaranteeing 100% financial transparency.

Our Tech Stack

- Ethereum: The underlying blockchain network where our records live.
- Solidity: Used to write the "Smart Contracts" (code that automatically releases funds when milestones are met).
- Hardhat: The backend framework used to compile and test our smart contracts.
- MetaMask: The crypto-wallet integration used to securely authorize and process transactions.

SMART CONTRACT ARCHITECTURE

Smart Contract Design (TenderSystem.sol)

We use Blockchain because it is immutable. Once a fund release is recorded, it is permanently locked. It can never be deleted, altered, or hidden, guaranteeing 100% financial transparency.

- Developed using Solidity on Ethereum (Sepolia Testnet)
- Core contract: TenderSystem
- Stores all bid data directly on blockchain (On-Chain Data)

What it Stores:

- Contractor address
- Bid amount
- Project details
- ML validation status

Security::

1. Only Oracle wallet can update ML validation result
2. Prevents unauthorized modification

BLOCKCHAIN EXECUTION FLOW

Step 1: Bid Submission
(Frontend → Blockchain)

- Contractor fills bid form on Next.js UI
- MetaMask opens → user signs transaction
- Bid data stored on blockchain
- BidSubmitted event is triggered

Step 2: ML Evaluation
(Blockchain → Backend)

- FastAPI backend listens to blockchain events
- Extracts bid data from blockchain
- Runs XGBoost ML model
- Determines whether bid is valid

Step 3: Oracle Update
(Backend → Blockchain)

- Backend acts as Custom First-Party Oracle
- Uses web3.py to call smart contract

Step 4: Contract Award
(Government)

- Government views only ML-validated bids
- Selects contractor
- Blockchain ensures tamper-proof validation

RESULTS & DISCUSSION

The project has successfully implemented major components of the system.

- Backend APIs for authentication, project management, and fund tracking are functional
- Role-based dashboards for government, contractor, and citizen are developed
- Dataset preparation for machine learning model is completed
- Core modules such as project creation, milestone tracking, and feedback system are working

CONTRIBUTION TO SUSTAINABLE DEVELOPMENT GOALS

InfraSync contributes to multiple UN Sustainable Development Goals:

- **SDG 9 (Industry, Innovation & Infrastructure):**
  Enhances infrastructure planning using modern technologies

- **SDG 16 (Peace, Justice & Strong Institutions):**
  Promotes transparency and accountability in fund management

- **SDG 11 (Sustainable Cities & Communities):**
  Improves monitoring of urban infrastructure projects

The system supports the development of smart and transparent cities.

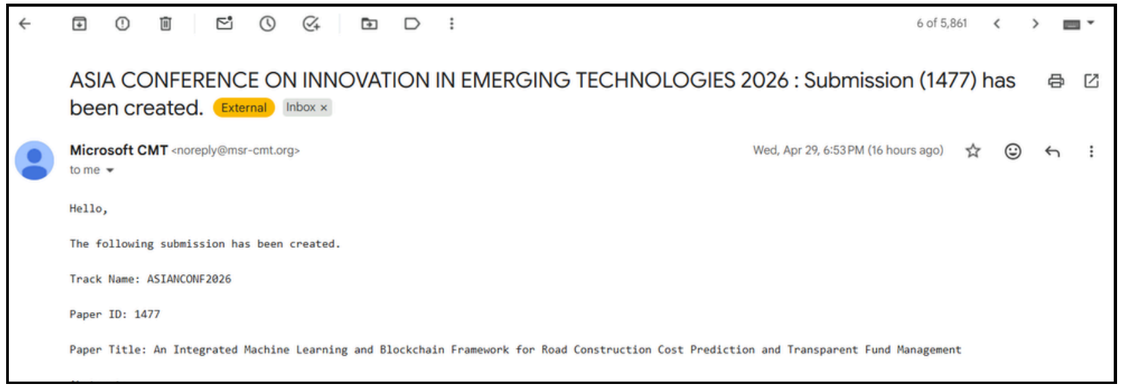
DETAILS OF PAPER PUBLICATION

Research Paper Title:
“An Integrated Machine Learning and Blockchain Framework for Road Construction Cost Prediction and Transparent Fund Management”

Authors
Om Sagar, Parag Pawar, Harshal Suroshe, Vinayak Suryawanshi, Prof. Rucha Shinde

Conference
Asia Conference on Innovation in Emerging Technologies 2026

Status
Paper Submitted Successfully



CONCLUSION & FUTURE SCOPE

Conclusion

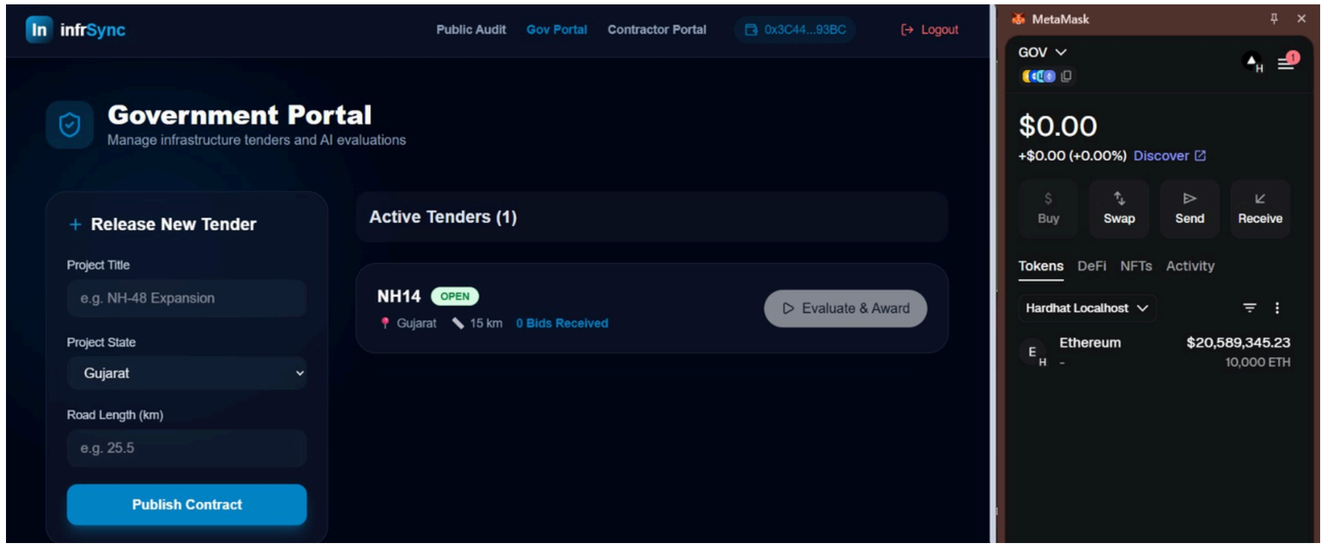
InfraSync provides an effective solution to improve transparency, planning, and monitoring of infrastructure projects. By combining Machine Learning and Blockchain, it enables better decision-making and reduces inefficiencies.

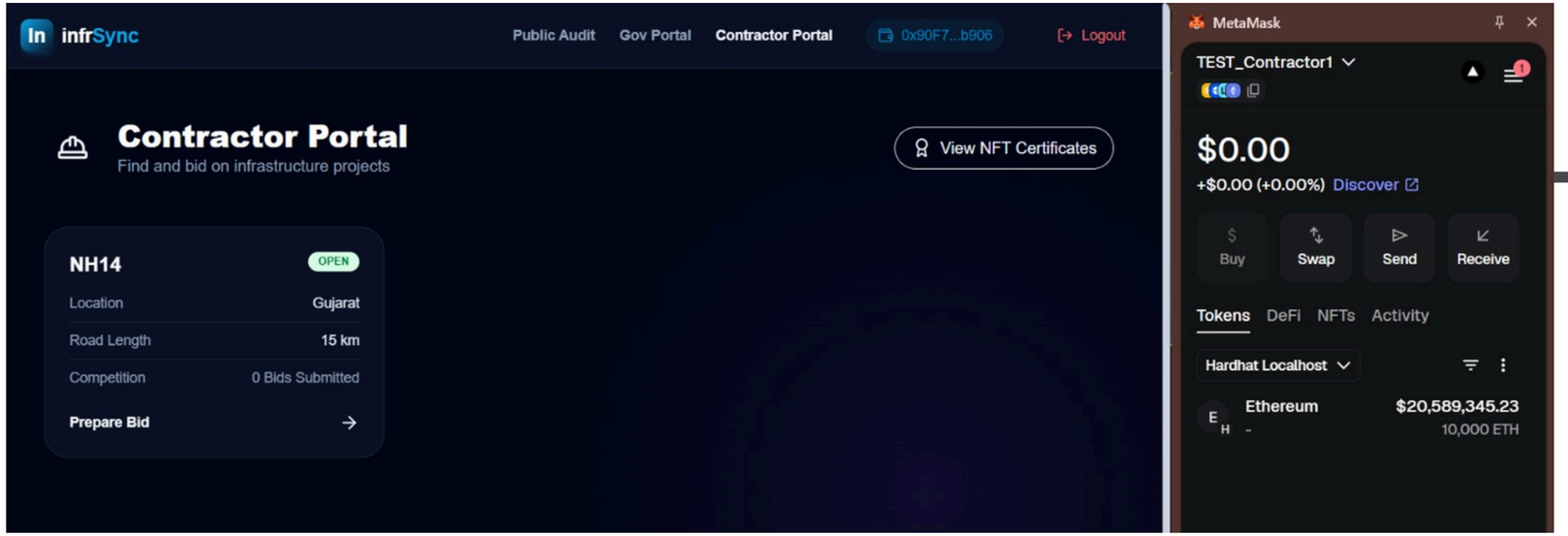
Future Scope

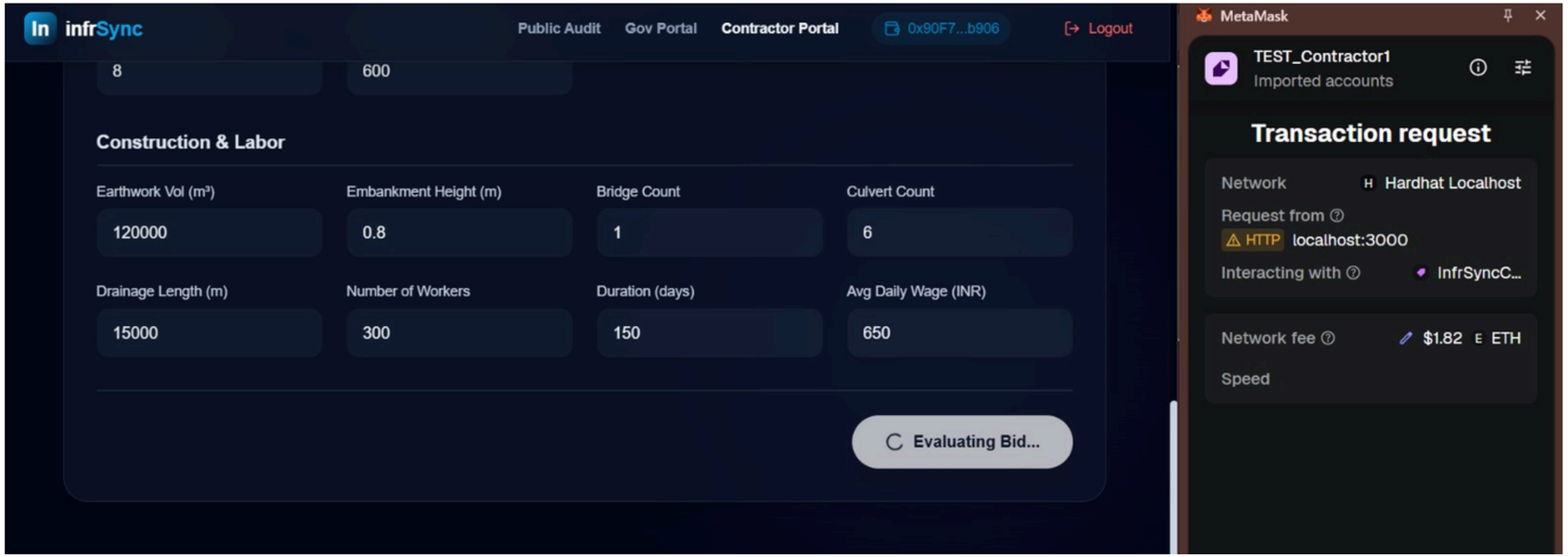
- Integration with real-world government systems
- Implementation of advanced ML models like XGBoost
- Mobile application for easy access
- Integration of IoT sensors for real-time road condition monitoring

The project has strong potential to evolve into a real-world smart governance solution.

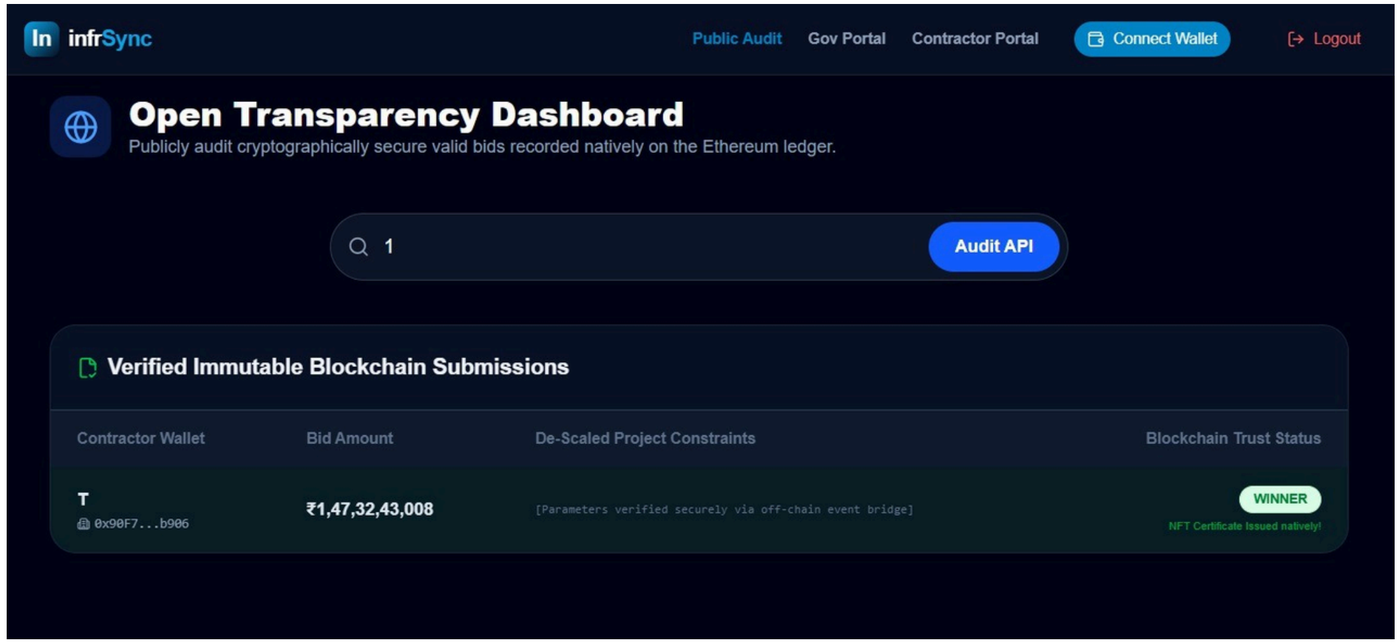
DASHBOARD SCREENSHOTS






infrSync
Public Audit
Gov Portal
Contractor Portal
0x90F7...b906
Logout
8
600
Construction & Labor
Earthwork Vol (m³)
120000
Embankment Height (m)
0.8
Bridge Count
1
Culvert Count
6
Drainage Length (m)
15000
Number of Workers
300
Duration (days)
150
Avg Daily Wage (INR)
650
Evaluating Bid...

MetaMask
TEST_Contractor1
Imported accounts
Transaction request
Network
Hardhat Localhost
Request from
HTTP
localhost:3000
Interacting with
InfrSyncC...
Network fee
$1.82
ETH
Speed


infrSync
Public Audit
Gov Portal
Contractor Portal
Connect Wallet
Logout
Open Transparency Dashboard
Publicly audit cryptographically secure valid bids recorded natively on the Ethereum ledger.
1
Audit API
Verified Immutable Blockchain Submissions
Contractor Wallet
Bid Amount
De-Scaled Project Constraints
Blockchain Trust Status
T
0x90F7...b906
₹1,47,32,43,008
[Parameters verified securely via off-chain event bridge]
WINNER
NFT Certificate Issued natively!

SYSTEM DESIGN & REQUIREMENT ANALYSIS

We designed three distinct portals based on user needs:



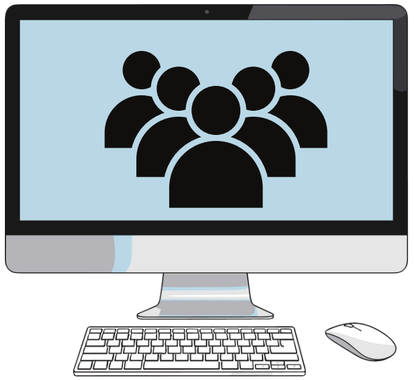
Government Portal

Central dashboard to create projects, allocate funds, and monitor Gantt-style timelines.



Contractor Portal

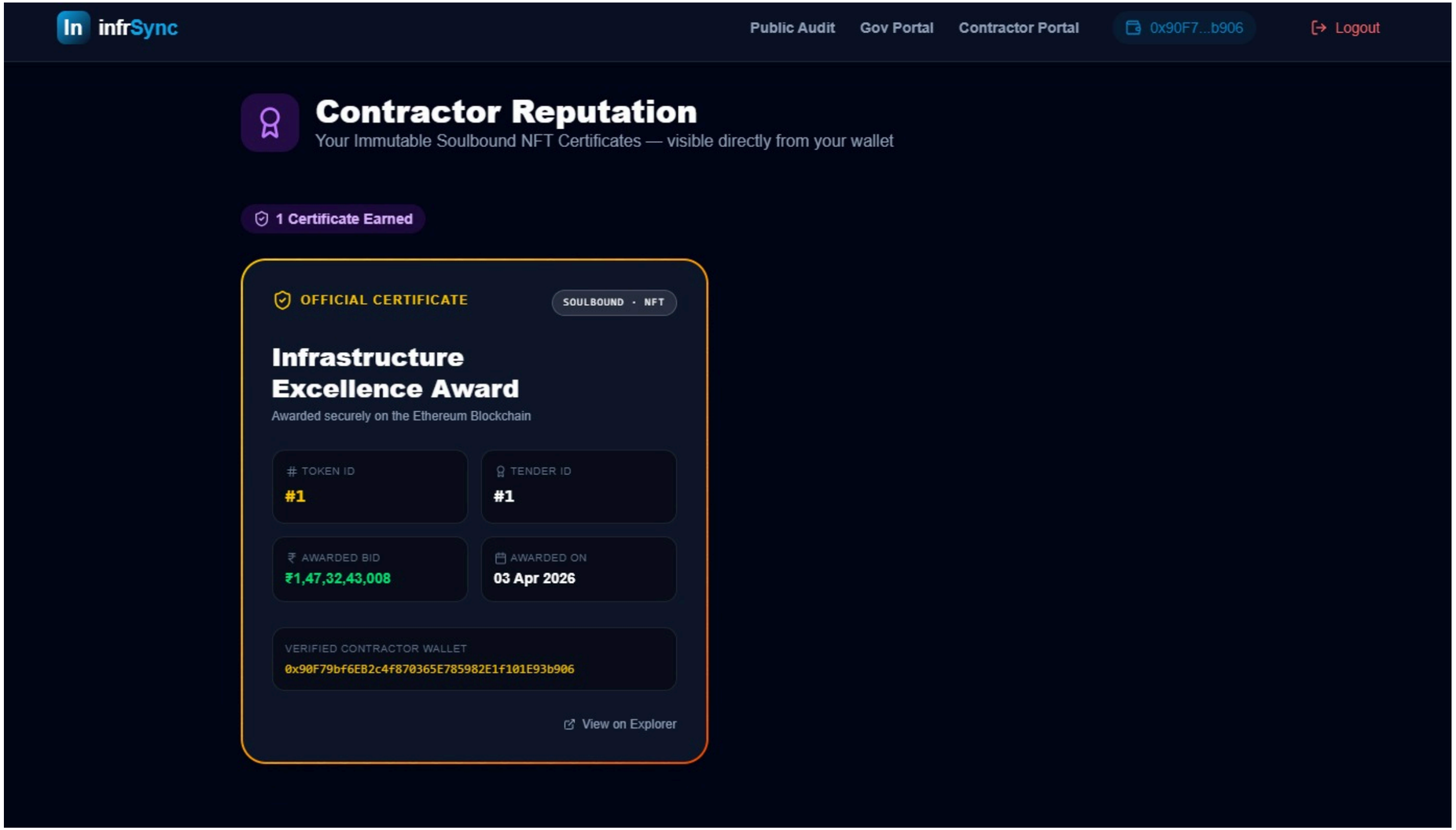
Workspace to view assigned projects, update milestones, and track payments.



Citizen Portal

Public interface to view nearby projects and submit feedback.

CERTIFICATE IMPLEMENTATION COMPLETED



Thank you!